10.06.2022

EE 2004 Microprocessor Systems
FINAL EXAM SOLUTION MANUAL

Name:

Q1) (20 points) The below assembly code represents a state machine. Please analyze it and plot the state diagram corresponding to the code.

```
    MOV    R0, #0x93
    LDR    R1, =0x20000040
    STR    R0, [R1]
    LDR    R2, [R1]

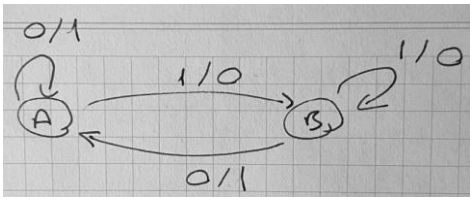
    MOV    R3, #0x01
    MOV    R5, #0x00
    MOV    R6, #0x06
    ANDS   R4, R2, R3
    LSR    R2, R2, #1
    CMP    R4, #0x0
    BEQ    ST_A_1
    BNE    ST_B_0

ST_A_0
    LSL    R5,R5, #1
    B      ST_A
ST_A_1
    ADD    R5,R5,#0x01
    LSL    R5,R5, #1
ST_A
    SUBS   R6,R6,#0x01
    BEQ    OUT
    LSR    R2, R2, #1
    ANDS   R4, R2, R3
    CMP    R4, #0x0
    BEQ    ST_A_1

ST_B_0
    LSL    R5,R5, #1
    B      ST_B
ST_B_1
    ADD    R5,R5,#0x01
    LSL    R5,R5, #1
ST_B
    SUBS   R6,R6,#0x01
    BEQ    OUT
    LSR    R2, R2, #1
    ANDS   R4, R2, R3
    CMP    R4, #0x0
    BEQ    ST_A_1
    B      ST_B_0

OUT
    STR R5, [R1,#0X04]
```

Solution:



Q2) (10 points) Please answer the below questions with at most one sentence for each part.

Please answer each part of the question in its designated place. Otherwise, your answer will not be graded.

a- UART stands for

b- The difference between the UART and USART is

c- If the data length is 100 bytes, then how many packets should be sent in UART communication? Here, data size is set as 8 bits in UART communication.

d- If the data transfer rate is set as 9600 bps, then how long does it take to transfer 100 bytes of data? You can take extra bits used in a packet, besides data, as 2 bits.

Solution:

a- UART stands for Universal Asynchronous Receiver-Transmitter.

b- The difference between the UART and USART is the clock signal used in USART.

c- If the data length is 100 bytes, then how many packets should be sent in UART communication? Here, data size is set as 8 bits in UART communication.

100 bytes = 800 bits.
Each packet can carry 8 bits. Hence, 800/8 = 100 packets should be sent.

d- If the data transfer rate is set as 9600 bps, then how long does it take to transfer 100 bytes of data? You can take extra bits used in a packet, besides 8-bit data, as 2 bits.

Each packet consists of 10 bits. 100 packets => 1000 bits. 1000/9600 sec is needed.

Q3) (30 points) Annual maintenance is important for a device. Therefore, an alarm should be displayed daily starting from 30 days before the designated maintenance day. Please design the alarm with the following constraints in mind. You should use Mbed Studio in designing the system.

a- The alarm should be displayed as the device is turned on every day. Turning on the device generates a port interrupt which can be sensed from the pin PC_13.
b- Real-time clock should be used in operation. Initially the time is set such that we have exactly one year for maintenance.
c- The alarm will be simulated by changing the brightness of the LED connected to the pin PA_5. The operation should be set such that if the date is outside the 30-day alarm period, then the LED is turned off. Within the 30-day alarm period, the brightness of the LED increases as the date becomes closer to the final day. On the final maintenance day, the LED will have the highest brightness.

Please answer each part of the question in its designated place below.
Otherwise, your answer will not be graded.

```
#include "mbed.h"

tm date;

//INITIALIZATIONS TO BE ADDED (5 points)
//------------------------------------------------------------------------//




//------------------------------------------------------------------------//

//CALLBACK FUNCTION TO BE ADDED (3 points)
//------------------------------------------------------------------------//



//------------------------------------------------------------------------//

int main(){
date.tm_year = 122;date.tm_mon = 5;date.tm_mday = 10;date.tm_hour = 14;
date.tm_min = 00;date.tm_sec = 00;
set_time(mktime(&date));

time_t initial=time(NULL);

//PORT INTERRUPT ADJUSTMENT (2 points)
//------------------------------------------------------------------------//




//------------------------------------------------------------------------//
```

```
//TIME BASED VARIABLE SETTINGS (2 points)
//-------------------------------------------------------------------//



//-------------------------------------------------------------------//

//SETTINGS TO BE USED IN ADJUSTING BRIGHTNESS OF THE LED (3 points)
//-------------------------------------------------------------------//


//-------------------------------------------------------------------//

while (true){

//PORT INTERRUPT AND TIME-BASED OPERATIONS (8 points)
//-------------------------------------------------------------------//









//-------------------------------------------------------------------//

//ADJUSTING BRIGHTNESS OF THE LED (7 points)
//-------------------------------------------------------------------//









//-------------------------------------------------------------------//
}}}
```

Solution:

```
#include "mbed.h"

tm date;

//INITIALIZATIONS TO BE ADDED (5 points)
//-------------------------------------------------------------------//
InterruptIn userButton(BUTTON1);
uint8_t flag = 0;
PwmOut pwmOut(LED1);
//-------------------------------------------------------------------//

//CALLBACK FUNCTION TO BE ADDED (3 points)
//-------------------------------------------------------------------//
void trigger(){
flag = 1;
}
//-------------------------------------------------------------------//

int main(){
date.tm_year = 122;date.tm_mon = 5;date.tm_mday = 10;date.tm_hour = 14;
date.tm_min = 00;date.tm_sec = 00;
set_time(mktime(&date));

time_t initial=time(NULL);

//PORT INTERRUPT ADJUSTMENT (2 points)
//-------------------------------------------------------------------//
userButton.rise(&trigger);
//-------------------------------------------------------------------//

//TIME-BASED VARIABLE SETTINGS (2 points)
//-------------------------------------------------------------------//
time_t target;
time_t days_remaining;
//-------------------------------------------------------------------//

//SETTINGS TO BE USED IN ADJUSTING THE BRIGHTNESS OF THE LED (3 points)
//-------------------------------------------------------------------//
pwmOut.period_us(256);
//-------------------------------------------------------------------//

while (true){

//PORT INTERRUPT AND TIME-BASED OPERATIONS (8 points)
//-------------------------------------------------------------------//
if (flag == 1){

time_t present = time(NULL);
```

```
printf("Initial Time = %d\r\n", initial);
printf("Present Time = %d\r\n", present);

target=initial+(365-30)*24*60*60;

//days_remaining=(target-present)/(24*60*60);
days_remaining=(target-present)/(24*60*60)-306;

if (days_remaining==0)
   initial=present;

printf("Days remaining = %d\r\n", days_remaining);
//------------------------------------------------------------------------//

//ADJUSTING BRIGHTNESS OF THE LED (7 points)
//------------------------------------------------------------------------//
if (days_remaining<30)
pwmOut.write((30.0-days_remaining)/30.0);
else
pwmOut.write(0.0);

flag = 0;
//------------------------------------------------------------------------//
}}}
```

Q4) (40 points) In this question, we will design a smart sensor to observe the effect of an earthquake on a building. There are two accelerometers available for your usage. The first accelerometer only detects the start time of the earthquake. It can generate a port interrupt accordingly. The first accelerometer is connected to the pin PC_13. The second accelerometer can measure the effect of the earthquake on the building in analog form. The second accelerometer is connected to the pin PC_12. Please follow the below constraints while designing your system. You should use Mbed Studio in designing the system.

Please answer each part of the question in the below designated place.
Otherwise, your answer will not be graded.

a- Generate a port interrupt operation, with all its settings, related functions, and parameters such that the timer starts working as the start of the earthquake is detected.

b- As the timer turns on, it generates 64 interrupts in one second. After 64 interrupts and all related operations are done, the timer is turned off. The system goes to low-power mode. Generate a timer interrupt operation, with all its settings, related functions, and parameters. Hint: 1/64 s = 15625 us.

c- At every timer interrupt, the analog accelerometer data is acquired by the ADC module. The ADC module has 12-bit resolution. Hence, the acquired data has range 0x0000 to 0x0FFF. This data should be stored in a 64-element array using pointer arithmetic. Generate ADC operation, with all its settings, related functions, and parameters.

d- The system goes to low power mode when not used.

Solution:

```
#include "mbed.h"

#define CONV_NUM 64

Ticker adcTrigger;
AnalogIn aIn(ADC_TEMP);
InterruptIn userButton(BUTTON1);

int cnt = 0;
bool adcFlag = false;

void ADCTrigger(){
adcFlag = true;
}

bool StartFlag = false;

void StartSensor(){
StartFlag = true;
adcTrigger.attach(&ADCTrigger, 15625us);
}

uint16_t value_arr[64];

int main(){
userButton.fall(&StartSensor);
__enable_irq();

while (true){
if (StartFlag){
printf("\n\n ADC Conversion Started \n");
StartFlag = false;
}

if (adcFlag && cnt < CONV_NUM){
value_arr[cnt] = aIn.read_u16();
 cnt++;
adcFlag = false;
}

if (cnt >= CONV_NUM){
adcTrigger.detach();
cnt=0;
printf("values= ");
for (int ind = 0; ind < CONV_NUM; ind++){
printf(" 0x%04X, ", value_arr[ind]);
}}
sleep();
}}
```